

Remainder Theorem

Dividing a polynomial $f(x)$ by $x - k$ leaves remainder $f(k)$.

$$f(x) = x^3 - 4x + 1$$

What is the remainder when $f(x)$ is divided by $x - 2$?

A negative test value

$$f(x) = 2x^3 + x^2 - 5$$

What is the remainder when dividing $f(x)$ by $x + 1$?

Factor Theorem

For a polynomial f , $x - k$ is a factor exactly when $f(k) = 0$.

$$f(x) = x^3 - 2x^2 - 5x + 6$$

Is $x - 1$ a factor?

Finding an unknown coefficient

$$f(x) = x^3 + ax + 6$$

What value of a makes $x - 2$ a factor?

Reducing the degree

The polynomial has a known zero at $x = 1$.

$$f(x) = x^3 - 2x^2 - 5x + 6$$

What quadratic factor remains after dividing by $x - 1$?

Completing the zero search

$$f(x) = (x - 1)(x^2 - x - 6)$$

What are all the zeros?

Rational Zero Theorem

For integer coefficients, a rational zero p/q in lowest terms has p dividing the constant and q dividing the leading coefficient.

Does the theorem guarantee that every candidate is a zero?

Listing rational candidates

$$f(x) = 2x^3 - 3x^2 - 8x + 12$$

What distinct rational-zero candidates does the theorem allow?

Testing candidates

$$f(x) = 2x^3 - 3x^2 - 8x + 12$$

Which positive candidates are zeros?

Finding all zeros

$$f(x) = 2x^3 - 3x^2 - 8x + 12$$

What are all zeros after using a confirmed rational factor?

No rational zeros

$$f(x) = x^2 - 2$$

Why does failure of the rational candidates not mean there are no real zeros?

Fundamental Theorem of Algebra

Every nonconstant polynomial has at least one complex zero.

Must a nonconstant polynomial have a real zero?

Counting complex zeros

A degree n polynomial has exactly n complex zeros when multiplicity is counted.

$$f(x) = (x - 2)^3(x + 1)^2$$

How many zeros are counted, and how many are distinct?

Nonreal conjugate pairs

For real coefficients, nonreal zeros occur in conjugate pairs.

If $2 + 3i$ is a zero, what other zero must occur?

Complex zeros

$$f(x) = (x - 3)(x^2 + 4)$$

What are all complex zeros?

Linear factorization

Over the complex numbers, a degree n polynomial factors into n linear factors, counting repeats.

$$f(x) = x^2 + 9$$

What is its linear factorization?

Building a polynomial

Find a monic polynomial with zeros -2 , 1 , and 4 .

What factored form has exactly these zeros?

Building with multiplicities

Find a monic polynomial with zero -1 of multiplicity 2 and zero 3 of multiplicity 1.

What polynomial satisfies these conditions?

Building from a complex zero

Find a monic quadratic with real coefficients and zero $1 + 2i$.

What is the quadratic in expanded form?

Descartes' Rule of Signs

The number of positive real zeros, counting multiplicity, equals the sign changes in $f(x)$ or is less by an even number.

$$f(x) = x^4 - 3x^3 + 2x^2 + x - 5$$

What positive-zero counts are possible?

Possible negative zeros

Apply Descartes' Rule of Signs to $f(-x)$.

$$f(x) = x^3 + 2x^2 - 5x - 6$$

What negative-zero counts are possible?

Using a graph to guide a test

$$f(x) = x^3 - x - 1$$

Which pair of consecutive integers brackets a real zero?

A volume equation

A box has width x cm, length $x + 2$ cm, and height $x - 1$ cm. Its volume is 30 cubic centimeters.

What polynomial equation and physical restriction describe x ?

A physical zero

For the box dimensions x , $x + 2$, and $x - 1$ cm:

$$x^3 + x^2 - 2x - 30 = 0$$

Which solution gives physically valid dimensions?